AI's Limits in Programming

1) AI still falls short on programming language design

Source: https://www.thenews.com.pk/latest/1403074-ai-still-falls-short-on-programming-language-design-bjarne-stroustrup-says

Bjarne Stroustrup, creator of the C++ programming language, says humans will still have the upper hand on programming.

Key takeaways:

- He believes that AI produces buggy, security issue prone code that's difficult to fix and review properly.

- Humans make localized changes. AI does not and can make changes in other places without telling the user.

- Senior developers do not want to deal with validating AI-generated code due to its unreadability

- AI can be unpredictable and cause a paradigm shift.

Overall, AI should not be relied on for such tasks, as it can be unpredictable and can especially cause issues due to this in designing programming languages.

2) State of AI vs human code generation report

Source: https://www.coderabbit.ai/blog/state-of-ai-vs-human-code-generation-report

CodeRabbit released a report suggesting that while AI increased output, it also increased the number of mistakes that had to be fixed by humans.

Key takeaways:

- Pull requests that were AI-generated had 1.7 times more issues compared to human-written pull requests

- Pull requests generated by AI were more likely to have critical bugs that would need to be fixed by humans later

- Logic issues were 75% more common in AI-generated pull requests

- AI lacked the ability to consistently make clean, easy to read code. This makes fixing mistakes or making changes to its code later on far more difficult than if it was written by a human.

- AI is more likely to forego error handling in the code it generates, making the features it would add more unstable.

- AI was 2.74 times more likely to introduce security issues into its code.

- Small performance issues were more likely to be caused by AI

- AI was ~2 times more likely to have inconsistent formatting and names in the code it generates

AI was more prone overall to causing instability in its code, meaning that it is not at the point at which it can replace humans.

3) Challenges and paths towards AI for Software Engineering

Source: https://arxiv.org/pdf/2503.22625

AI still has many challenges to overcome in the software engineering field.

- AI suffers from contamination, caused by being trained on its own data from the internet

- AI contamination can cause LLMs to degrade in quality as they get subjected to more low-quality, poorly written AI code

- LLMs struggle with security configurations, mixing up Task Execution Roles with Task Roles.

- AI hallucinations can affect the user's idea of how the code should work, leading to more mistakes and potential code rewrites.

- Pull request reviews can be automated easily

- AI struggles with fixing bugs in more complex and larger codebases

- Humans should review security issues instead, since it's easier for them to spot such issues.

- Porting code to another programming language should involve lots of human oversight, as AI can cause safety issues.

Overall, AI needs to be looked over by humans to make sure it does not introduce critical issues. Although it can be used to automate some things, AI can hallucinate and introduce bugs that would need human review to fix.